

Interview Readiness Analyzer

Machine Learning Evaluation Report

Model : XGBoost Regressor

Dataset Size : 1000

Evaluation Metrics

MAE : 2.81

MSE : 12.1

RMSE : 3.48

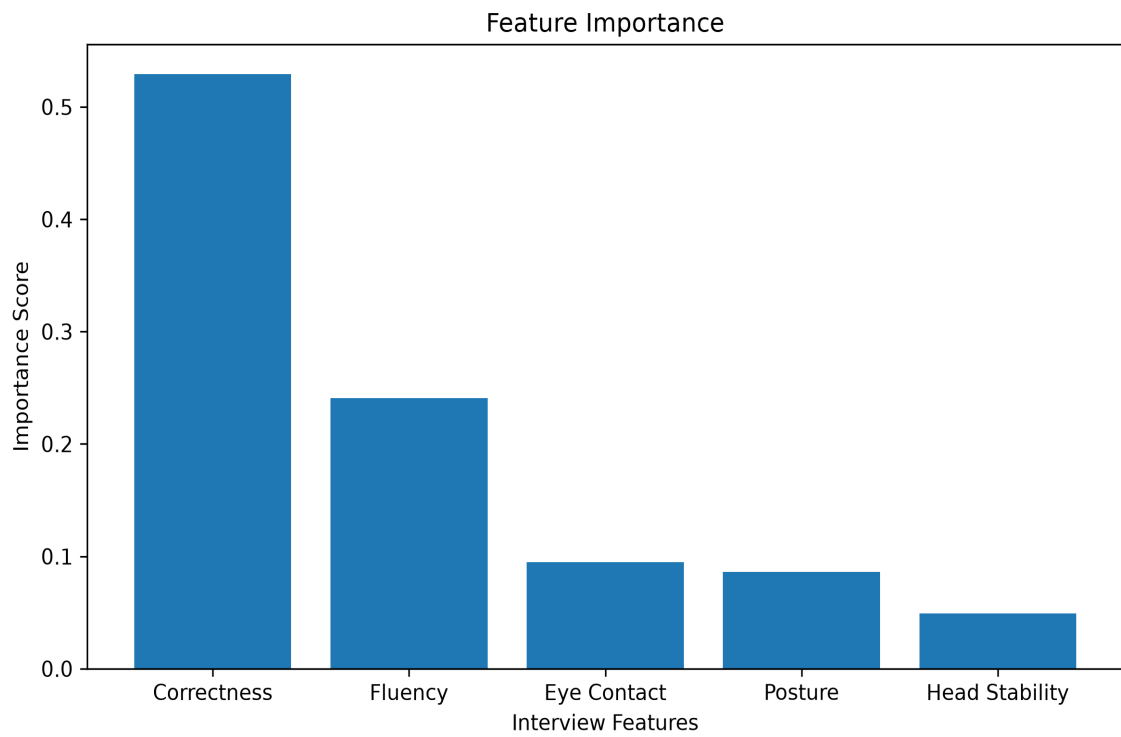
R² Score : 0.8523

Best Hyperparameters

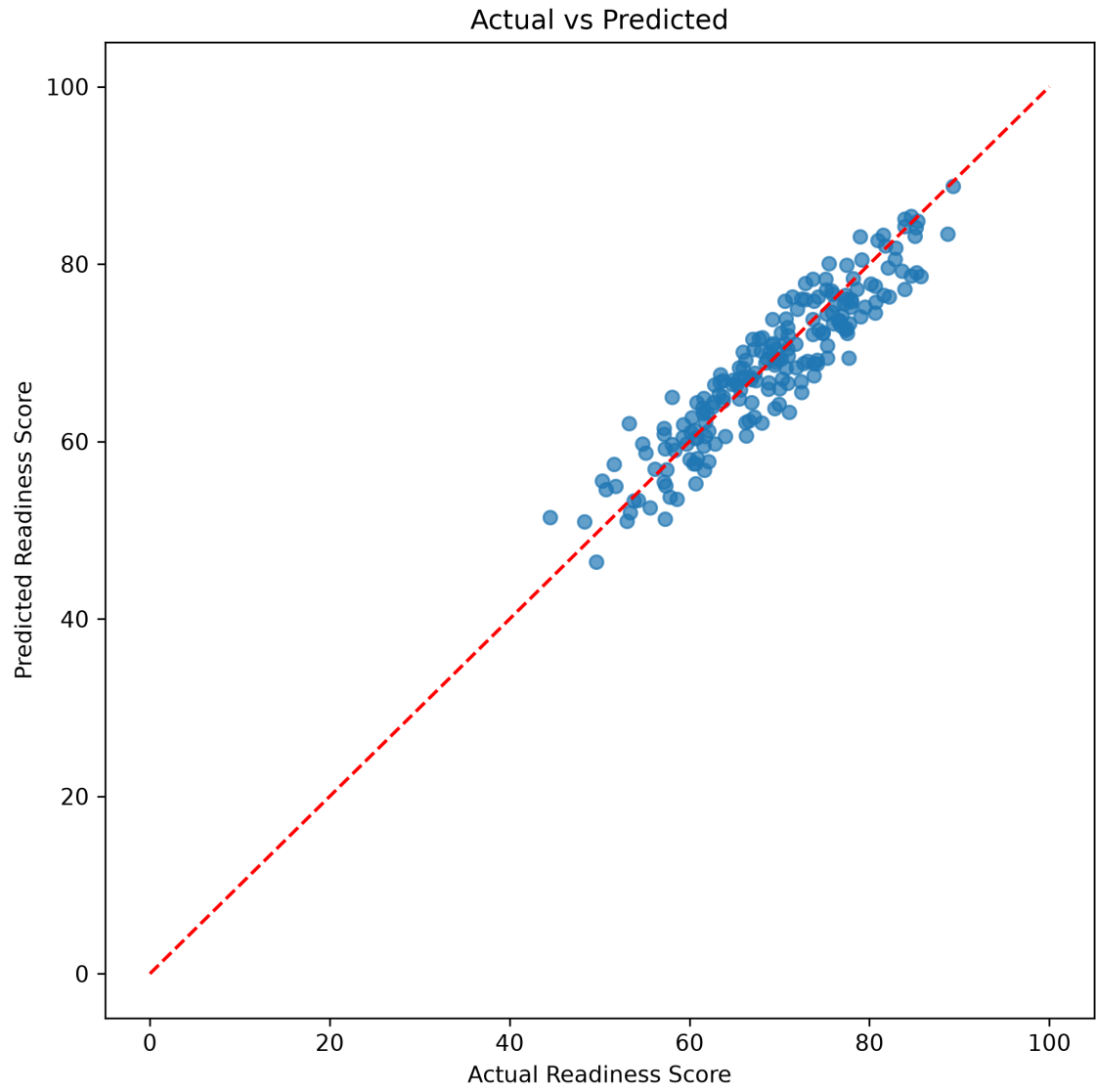
objective : reg:squarederror
base_score : None
booster : None
callbacks : None
colsample_bylevel : None
colsample_bynode : None
colsample_bytree : 0.8
device : None
early_stopping_rounds : None
enable_categorical : True
eval_metric : None
feature_types : None
feature_weights : None
gamma : None
grow_policy : None
importance_type : None
interaction_constraints : None
learning_rate : 0.05
max_bin : None
max_cat_threshold : None
max_cat_to_onehot : None
max_delta_step : None
max_depth : 6
max_leaves : None
min_child_weight : None
missing : nan
monotone_constraints : None
multi_strategy : None
n_estimators : 300
n_jobs : None
num_parallel_tree : None
random_state : 42
reg_alpha : None
reg_lambda : None
sampling_method : None
scale_pos_weight : None

subsample : 0.8
tree_method : None
validate_parameters : None
verbosity : None

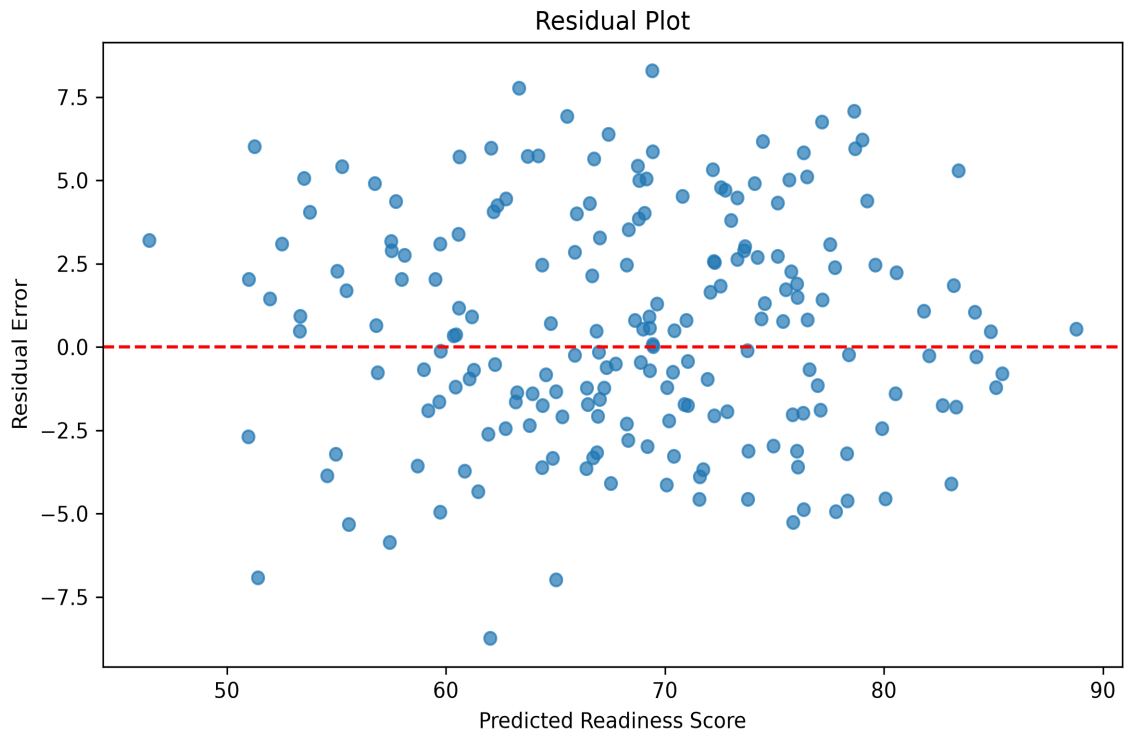
Feature Importance



Actual vs Predicted



Residual Plot



Model Interpretation

The XGBoost model achieved a strong R^2 score of approximately 0.86, indicating that it successfully explains most of the variation in candidate interview readiness. Feature importance analysis shows that technical correctness contributes the most to the final prediction, followed by fluency, eye contact, posture and head stability. The Actual vs Predicted plot demonstrates good agreement between predicted and true readiness scores, while the residual plot shows randomly distributed errors around zero, indicating that the model does not suffer from systematic prediction bias. Overall, the trained model is suitable for deployment within the Interview Readiness Analyzer.